

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - ANSWER SCRIPT

Course Name / Code: CSA16 - Data Warehousing and Data Mining

Course Outcome: CO2 - Apply suitable Data Preprocessing techniques (BL3, BL4)

Student Name: M. Balaji

Register Number: 192565072

Declaration: I, M. Balaji (Reg No: 192565072), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Q1. Data Preprocessing in Retail Analytics

A. Handling Missing Values

(i) Missing values identified: Age is missing for customer C2, Monthly_Spend is missing for C3, and Purchase_Frequency is missing for C5.

(ii)-(iii) Imputation - Method 1: Median Imputation. The median is used because it is robust to the extreme values already present in the data (e.g. Age = 105, Spend = 500), unlike the mean which such outliers would distort.

- Age (sorted, excluding C2): 23, 27, 31, 36, 40, 42, 105 → Median = **36**. Imputed Age for C2 = 36.
- Monthly_Spend (sorted, excluding C3): 20, 25, 35, 38, 45, 300, 500 → Median = **38**. Imputed Spend for C3 = 38.
- Purchase_Frequency (sorted, excluding C5): 5, 8, 9, 10, 12, 45, 50 → Median = **10**. Imputed Frequency for C5 = 10.

Method 2: k-NN Imputation (k = 3). Distance is computed using the other complete numeric attributes of each record (Euclidean distance), and the missing value is estimated as the average of the 3 nearest neighbours. This captures local similarity rather than a single global statistic.

- C2 (Age missing) - using Spend & Frequency (35, 8) as coordinates, the 3 nearest neighbours are C7 (36,9), C6 (40,12) and C1 (23,5). Average of their Ages = $(36+40+23)/3 = \mathbf{33}$.
- C3 (Spend missing) - using Age & Frequency (31, 10), nearest neighbours are C7, C6, C1. Average of their Spend = $(38+45+20)/3 = \mathbf{34.3}$.
- C5 (Frequency missing) - using Age & Spend (27, 25), nearest neighbours are C1, C7, C6. Average of their Frequency = $(5+9+12)/3 = \mathbf{8.7}$.

Justification: Median imputation is simple, fast and outlier-resistant - suitable as a quick baseline. k-NN imputation is preferred here because it leverages the similarity between customers (e.g. similar spend/frequency behaviour) and therefore gives a more context-aware, realistic estimate for a recommendation system, even though it is more computationally expensive.

B. Outlier Detection and Treatment (Z-score method)

Using the median-imputed dataset, $Z = (X - \text{mean}) / \text{std.dev.}$ For **Age**: mean = 42.5, std.dev = 24.37. For **Monthly_Spend**: mean = 125.13, std.dev = 166.55.

Customer	Attribute	Z-score calculation	Result
C4	Age = 105	$(105-42.5)/24.37 = 2.56$	Outlier ($ Z \geq 2$)
C4	Spend = 500	$(500-125.13)/166.55 = 2.25$	Outlier ($ Z \geq 2$)

Customer	Attribute	Z-score calculation	Result
C8	Spend = 300	$(300-125.13)/166.55 = 1.05$	Not an outlier

Extreme records: C4 (Age = 105, Spend = 500) is confirmed as a genuine outlier on both attributes - Age = 105 is not realistically possible and Spend = 500 (i.e. ■5,00,000) is far beyond the normal range, most likely a data-entry error.

Treatment: (a) *Capping/Winsorizing* - cap Age at a realistic upper bound (e.g. 70) and cap Spend at mean + 2*std.dev ≈ 458 ; this retains the record instead of deleting it, which preserves sample size. (b) *Log transformation* could alternatively be applied to Monthly_Spend to compress the right-skew caused by high-value entries (C4, C8) while keeping all records. Removal is not recommended here since the dataset is already small (8 records) and removing rows would lose useful information from other attributes of C4.

C. Categorical Data Standardization

(i) Inconsistencies found in Gender: 'M', 'Male', 'male' all represent the same category; 'F', 'Female', 'FEMALE' all represent the same category - differing only in case and abbreviation.

(ii) Standardized mapping (Male / Female):

Customer	Standardized Gender
C1	Male
C2	Male
C3	Female
C4	Female
C5	Male
C6	Female
C7	Male
C8	Female

Q2. Covariance & Data Reduction in E-commerce Analytics

A. Covariance Analysis (Income vs Purchase_Value)

Mean(Income) = $645/8 = 80.625$; Mean(Purchase_Value) = $1590/8 = 198.75$. Covariance = $\sum(x-x\bar{x})(y-y\bar{y}) / n$.

Cust	Income	Purchase Value	$(x-x\bar{x})$	$(y-y\bar{y})$	$(x-x\bar{x})(y-y\bar{y})$
U1	40	100	-40.625	-98.75	4011.72
U2	60	150	-20.625	-48.75	1005.47
U3	85	210	4.375	11.25	49.22
U4	120	320	39.375	121.25	4774.22
U5	45	110	-35.625	-88.75	3161.72
U6	75	180	-5.625	-18.75	105.47
U7	150	360	69.375	161.25	11186.72
U8	70	160	-10.625	-38.75	411.72

Sum of products = 24706.26. Covariance = $24706.26 / 8 = 3088.3$ (population) [or $24706.26/7 = 3529.5$ as a sample covariance].

(ii) Interpretation: The covariance is strongly positive, meaning Income and Purchase_Value move together - customers with higher income tend to make higher-value purchases.

(iii) Role of covariance: Covariance quantifies the direction (and, together with the variables' scales, the strength) of the linear relationship between two features. A large positive or negative covariance signals redundancy - such features carry overlapping information and are candidates for removal or combination during feature selection/dimensionality reduction.

B. Feature Selection

(i) Redundant attributes: Income, Purchase_Value, Credit_Score, Orders, Avg_Balance and EMI all increase together across customers (U1→U7 shows a consistent upward trend in every column), indicating strong pairwise correlation and redundancy among these financial/behavioural attributes.

(ii) Optimal feature subset: {Income, Orders, Credit_Score} - retained as representative, less-redundant predictors of customer value.

(iii) Justification: Income captures overall purchasing power, Orders captures engagement frequency, and Credit_Score captures financial reliability - together they explain most of the variance needed to classify Customer_Class without duplicating information already carried by the highly-correlated Purchase_Value, Avg_Balance and EMI attributes. Dropping the redundant attributes reduces computational cost and the risk of multicollinearity in the predictive model.

C. Data Transformation (Standardization)

$Z = (X - \text{mean}) / \text{std.dev}$ is applied to every numeric column except Cust_ID and Customer_Class. Worked example for **Income**: mean = 80.625, std.dev = 35.13.

Customer	Income	Standardized Z-value
U1	40	-1.16
U2	60	-0.59
U3	85	0.12
U4	120	1.12
U5	45	-1.01

Customer	Income	Standardized Z-value
U6	75	-0.16
U7	150	1.98
U8	70	-0.30

The same procedure (subtract the column mean, divide by the column std.dev) is repeated for Age, Purchase_Value, Credit_Score, Orders, Avg_Balance and EMI.

(ii) Why standardize: The attributes are on very different scales and units - Credit_Score ranges in the hundreds, EMI in thousands of rupees, Orders in tens. Distance-based and gradient-based algorithms would otherwise let large-magnitude attributes (like EMI) dominate the model. Standardization rescales every attribute to mean 0 and standard deviation 1, ensuring each attribute contributes fairly to the model.

Q3. Data Transformation & Discretization in Education Analytics

A. Equal-Width Binning (3 bins)

Study_Hours: range = $11 - 2 = 9$, bin width = $9/3 = 3$. Bins: [2-5] Low, [5-8] Medium, [8-11] High.

- Low (2-4): S1(2), S2(3), S3(4)
- Medium (5-7): S4(5), S5(6), S6(7)
- High (8-11): S7(8), S8(9), S9(10), S10(11)

Test_Score: range = $100 - 40 = 60$, bin width = $60/3 = 20$. Bins: [40-60) Poor, [60-80) Average, [80-100] Excellent.

- Poor (40-59): S1(40), S2(50), S3(55)
- Average (60-79): S4(65), S5(75)
- Excellent (80-100): S6(85), S7(90), S8(95), S9(98), S10(100)

B. Equal-Frequency Binning (3 bins, ~equal record counts)

The 10 sorted records are split into groups of roughly equal size (3, 3, 4):

- Bin 1: S1, S2, S3 (Study_Hours 2-4 / Test_Score 40-55)
- Bin 2: S4, S5, S6 (Study_Hours 5-7 / Test_Score 65-85)
- Bin 3: S7, S8, S9, S10 (Study_Hours 8-11 / Test_Score 90-100)

Because the data here happens to already be sorted and evenly spaced, equal-frequency binning gives almost the same grouping as equal-width binning in this case; in general the two methods diverge when the data is skewed.

C. Concept Hierarchy Construction

- **Study_Hours** → Low (2-4 hrs), Medium (5-7 hrs), High (8-11 hrs)
- **Test_Score** → Poor (40-59), Average (60-79), Excellent (80-100)

This hierarchy converts the raw numeric attributes into interpretable categorical levels that align with the target Performance label, making downstream decision rules easier to explain.

Q4. Normalization of {200, 300, 400, 600, 1000}

(a) Min-Max Normalization (min = 0, max = 1)

$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$, where $X_{\min} = 200$, $X_{\max} = 1000$, range = 800.

X	Calculation	X' (Min-Max)
200	$(200-200)/800$	0.000
300	$(300-200)/800$	0.125
400	$(400-200)/800$	0.250
600	$(600-200)/800$	0.500
1000	$(1000-200)/800$	1.000

(b) Z-score Normalization

Mean = $(200+300+400+600+1000)/5 = 500$. Variance = $\Sigma(X-\text{mean})^2/n = 400000/5 = 80000$. Std.dev = $\sqrt{80000} = 282.84$.

X	Calculation	Z
200	$(200-500)/282.84$	-1.06
300	$(300-500)/282.84$	-0.71

X	Calculation	Z
400	$(400-500)/282.84$	-0.35
600	$(600-500)/282.84$	0.35
1000	$(1000-500)/282.84$	1.77

Q5. Data Transformation in Online Shopping Behavior Analysis

A. Equal-Frequency Binning (3 bins)

The 10 sorted durations are split into near-equal groups (3, 3, 4):

- **Bin 1 (Short):** C1(15), C2(30), C3(45)
- **Bin 2 (Medium):** C4(60), C5(120), C6(300)
- **Bin 3 (Long):** C7(600), C8(900), C9(1800), C10(3600)

Effectiveness: Equal-frequency binning groups the genuinely short visits (15-45s) well, but Bin 3 spans an enormous range (600 to 3600 seconds), lumping moderately long and extremely long visits together. This shows the method preserves record count per bin but does not by itself handle the strong right-skew - a customer who stayed 10 minutes and one who stayed an hour both land in the same 'long' bucket.

B. Z-Score Normalization

Mean = $7470/10 = 747$ seconds. Variance = $\Sigma(X-\text{mean})^2/n = 11901060/10 = 1190106$. Std.dev = $\sqrt{1190106} \approx 1090.9$.

Visit_Duration (s)	(X-mean)	Z-score
15	-732	-0.671
30	-717	-0.657
45	-702	-0.644
60	-687	-0.630
120	-627	-0.575
300	-447	-0.410
600	-147	-0.135
900	153	0.140
1800	1053	0.965
3600	2853	2.616

Handling extreme values: Z-score normalization does not bound the output to a fixed range, so extreme durations (1800s, 3600s) surface clearly as high positive Z-scores (0.97 and 2.62), correctly flagging them as unusually engaged customers. However, because these extreme values inflate the mean and standard deviation, they also compress the Z-scores of the shorter, more typical visits closer together.

C. Min-Max Normalization ([0, 1])

$X_{\min} = 15$, $X_{\max} = 3600$, range = 3585.

Visit_Duration (s)	X' (Min-Max)
15	0.0000
30	0.0042
45	0.0084
60	0.0126
120	0.0293
300	0.0795
600	0.1632
900	0.2469
1800	0.4979

Visit_Duration (s)	X' (Min-Max)
3600	1.0000

Comparison: Min-Max normalization squeezes almost all of the short-to-medium visits (15s-600s) into a very narrow band near 0 (0.000-0.163) because the single extreme value (3600s) stretches the whole scale, making these customers look artificially similar. Z-score normalization spreads the same short/medium values more evenly (roughly -0.67 to -0.14) because it centers on the mean rather than the range. In short, Min-Max is more sensitive to outliers since it is anchored directly to $X_{\min}/X_{\max}$, whereas Z-score normalization is comparatively more robust (though still affected) since it is based on the mean and standard deviation of the whole distribution. For this highly-skewed engagement data, Z-score (or a log-transform before scaling) is the better choice for clustering.

--- End of Answer Script ---